

13. (2.75 points)

An insurer uses the following information in an over-dispersed Poisson (ODP) bootstrap process by building a generalized linear model (GLM) based on incremental paid loss.

Cumulative Reported Loss (\$000) as of (months)			
Accident Year	12	24	36
2015	180	230	235
2016	200	180	
2017	225		

Open Claims as of (months)			
Accident Year	12	24	36
2015	3	1	0
2016	5	3	
2017	6		

Case reserves are set based on the maturity of the claim:

- 12 months = \$15,000 per claim
- 24 months = \$10,000 per claim
- 36 months = \$0 per claim.

a. (0.5 point)

Briefly describe two practical limitations of a log-link GLM bootstrap modeling framework.

b. (1.25 points)

Construct a log-link triangle that can be used to solve for the GLM parameters.

c. (1 point)

Calculate the fitted incremental paid results of the GLM model using the following parameters:

- $\alpha_1 = 5.04$
- $\alpha_2 = 4.66$
- $\alpha_3 = 4.91$
- $\beta_2 = -0.86$
- $\beta_3 = -1.47$

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 13

TOTAL POINT VALUE: 2.75

LEARNING OBJECTIVE(S): A7, A8

SAMPLE ANSWERS

Part a: 0.5 point

Sample responses:

- It cannot handle negative incremental values
- It cannot handle incomplete triangle
- Must be solved at each iteration making it time consuming
- GLM bootstrap requires computation of parameters at every step which can be computationally intensive
- Heteroscedasticity – when residuals don't have similar variance and they are not adjusted, GLM won't be appropriate or produce unstable results
- Make iid assumption in bootstrapping which won't work if non-constant variance across triangle
- Not easily explained to non-technical audience
- Sometimes the sum of residuals does not equal zero
- When the entire column sum of incremental losses is negative the GLM again will fail
- Can't incorporate CY trend
- ODP bootstrap GLM uses large number of parameters and may overfit data
- Needs adjustment for growth/shrinkage in book
- It cannot incorporate the deductibles and the policy limits as variable
- May underestimate variability if underlying triangle is limited & does not exhibit enough variability, as is the case with a small triangle

Part b: 1.25 points

Sample 1

Case Reserve Triangle

AY	12	24	36
2015	15 × 3=45	10	0
2016	75	30	
2017	90		
	↑ 15K per open claim	↑ 10K per open claim	

Open count × case reserve for each maturity

Cum Paid Triangle

AY	12	24	36
2015	135	220	235
2016	125	150	
2017	135		

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Incremental Paid Triangle

AY	12	24	36
2015	135	85	15
2016	125	25	
2017	135		

Take log of each cell

Log of Incremental Paid

AY	12	24	36
2015	4.91	4.44	2.71
2016	4.83	3.22	
2017	4.91		

Sample 2

Reported – # Open × Reserve = Paid

Incremental Paid

AY	12	24	36
2015	135,000	85,000	15,000
2016	125,000	25,000	
2017	135,000		

AY	12	24	36
2015	11.813	11.350	9.616
2016	11.736	10.127	
2017	11.813		

← Log(15,000)

Part c: 1 point

Sample 1

$$\ln q_{11} = \alpha_1 = 5.04$$

$$\ln q_{21} = \alpha_2 = 4.66$$

$$\ln q_{31} = \alpha_3 = 4.91$$

$$\ln q_{12} = \alpha_1 + \beta_2 = 4.18$$

$$\ln q_{22} = \alpha_1 + \beta_2 = 3.8$$

$$\ln q_{13} = \alpha_1 + \beta_2 + \beta_3 = 2.71$$

$$q_{11} = 154$$

$$q_{21} = 106$$

$$q_{31} = 136$$

$$q_{12} = 65$$

$$q_{22} = 45$$

$$q_{13} = 15$$

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Fitted incremental paid loss

AY	12	24	36
2015	154	65	15
2016	106	45	
2017	136		

Sample 2

	e^{α_1}	$e^{\alpha_1 + \beta_2}$	$e^{\alpha_1 + \beta_2 + \beta_3}$
AY	12	24	36
2015	154,500	65,400	15,000
2016	106,600	44,700	
2017	135,600		

EXAMINER'S REPORT

Candidates were expected to know background on the log-link GLM bootstrap modeling framework, including knowing how to construct a log-link triangle, and how to calculate fitted incremental paid results.

Part a

Candidates were expected to know practical limitations of log-link GLM bootstrap modeling framework.

Common mistakes included:

- Only providing one response.

Part b

Candidates were expected to construct a log-link triangle that can be used to solve for the GLM parameters.

Common mistakes included:

- Basing the calculation on incremental reported losses.

Part c

Candidates were expected to be able to calculate the fitted incremental paid results of the GLM model using given parameters.

Common mistakes included:

- Using an incorrect equation.